Hodgkin-Huxley Model

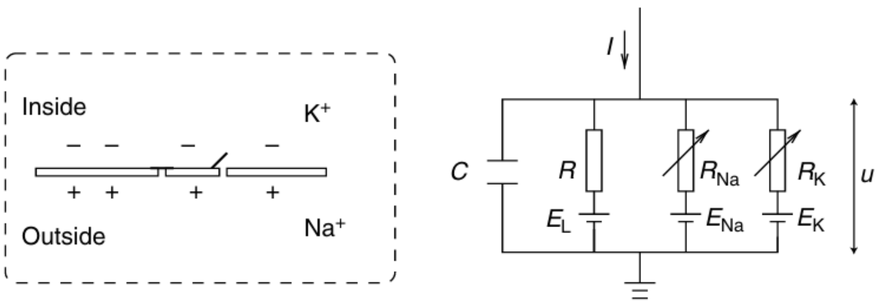
This model was created by British scientists in 1952.

The most impressive idea was intended to explain other ion channels in a very economical and straightforward way, making the model less prone to errors.

Introduction

The Hodgkin–Huxley model can be understood with the help of Fig. 2.2. The semiperme able cell membrane separates the interior of the cell from the extracellular liquid and acts as a capacitor. If an input current I(t) is injected into the cell, it may add further charge on the capacitor, or leak through the channels in the cell membrane. Each channel type is rep resented in Fig. 2.2 by a resistor. The unspecific channel has a leak resistance R, the sodium channel a resistance RNa and the potassium channel a resistance RK. The diagonal arrow across the diagram of the resistor indicates that the value of the resistance is not fixed, but changes depending on whether the ion channel is open or closed. Because of active ion transport through the cell membrane, the ion concentration inside the cell is different from that in the extracellular liquid. The Nernst potential generated by the difference in ion con centration is represented by a battery in Fig. 2.2. Since the Nernst potential is different for each ion type, there are separate batteries for sodium, potassium, and the unspecific third channel, with battery voltages ENa, EK and EL, respectively. Let us now translate the above schema of an electrical circuit into mathematical equa tions. The conservation of electric charge on a piece of membrane implies that the applied current I(t) may be split into a capacitive current IC which charges the capacitor C and further components Ik which pass through the ion channels. Thus where the sum runs over all ion channels. In the standard Hodgkin–Huxley model there are only three types of channel: a sodium channel with index Na, a potassium channel with index K and an unspecific leakage channel with resistance R; see Fig. 2.2. From the definition of a capacity C = q/u where q is a charge and u the voltage across the capacitor, we find the charging current IC =Cdu/dt.

2.2 Hodgkin–Huxley model



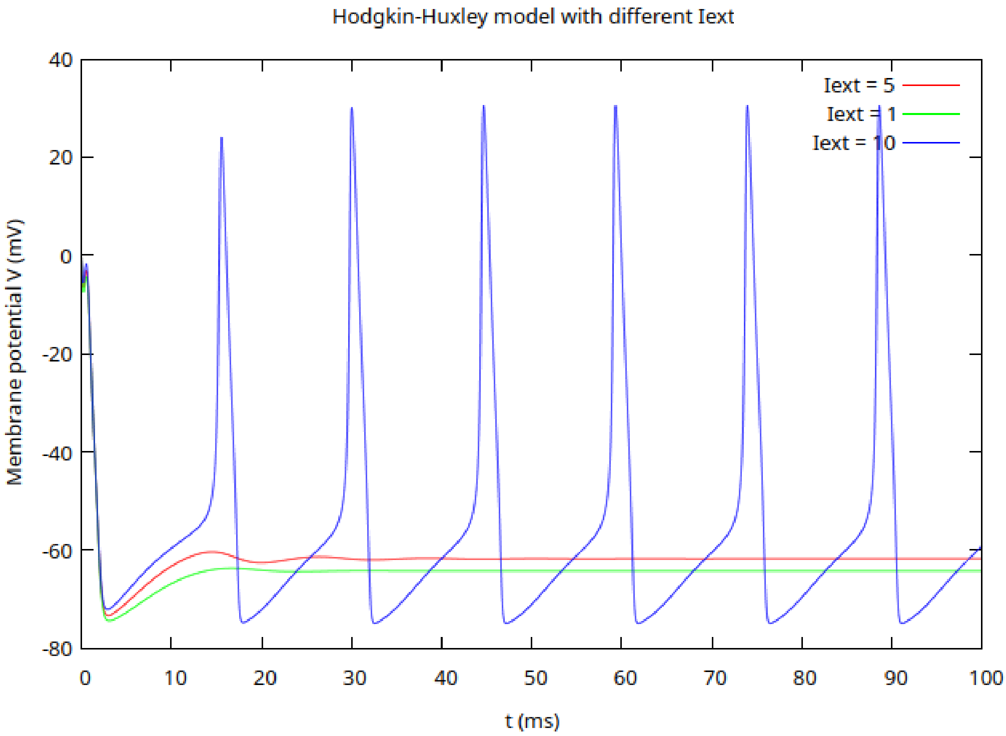
Constant (1)	Value chosen (2)
C_M (μF/cm^2)	1.0
V_{Na} (mV)	−115
V_K (mV)	+ 12
V_l (mV)	−10.613*
$\bar{g}_{Na}$ (m.mho/cm^2)	120
$\bar{g}_K$ (m.mho/cm^2)	36
$\bar{g}_l$ (m.mho/cm^2)	0.3

Table 3 from Hodgkin & Huxley showing empirical values for voltages and conductances, as well as the capacitance of the membrane.

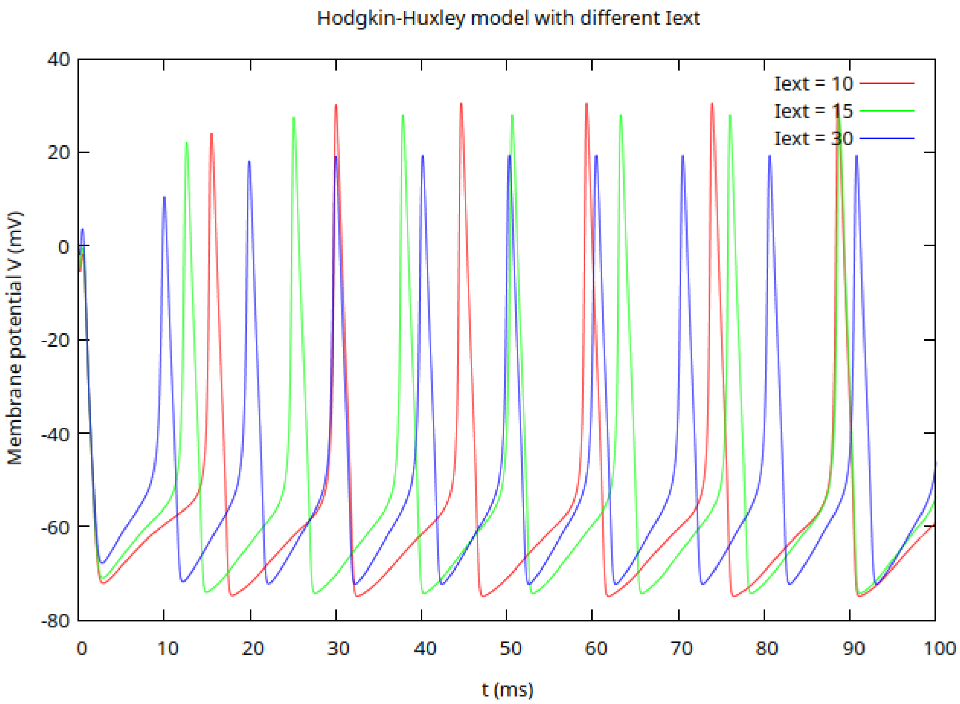
$$\begin{aligned}
\alpha_n &= 0.01\,(V+10)\Big/\left(\exp\frac{V+10}{10}-1\right),\\
\beta_n &= 0.125\,\exp(V/80),\\
\alpha_m &= 0.1\,(V+25)\Big/\left(\exp\frac{V+25}{10}-1\right),\\
\beta_m &= 4\,\exp(V/18),\\
\alpha_h &= 0.07\,\exp(V/20),\\
\beta_h &= 1\Big/\left(\exp\frac{V+30}{10}+1\right).
\end{aligned}$$

H & H's formulas for the n, m, and h gates as a function of voltage.

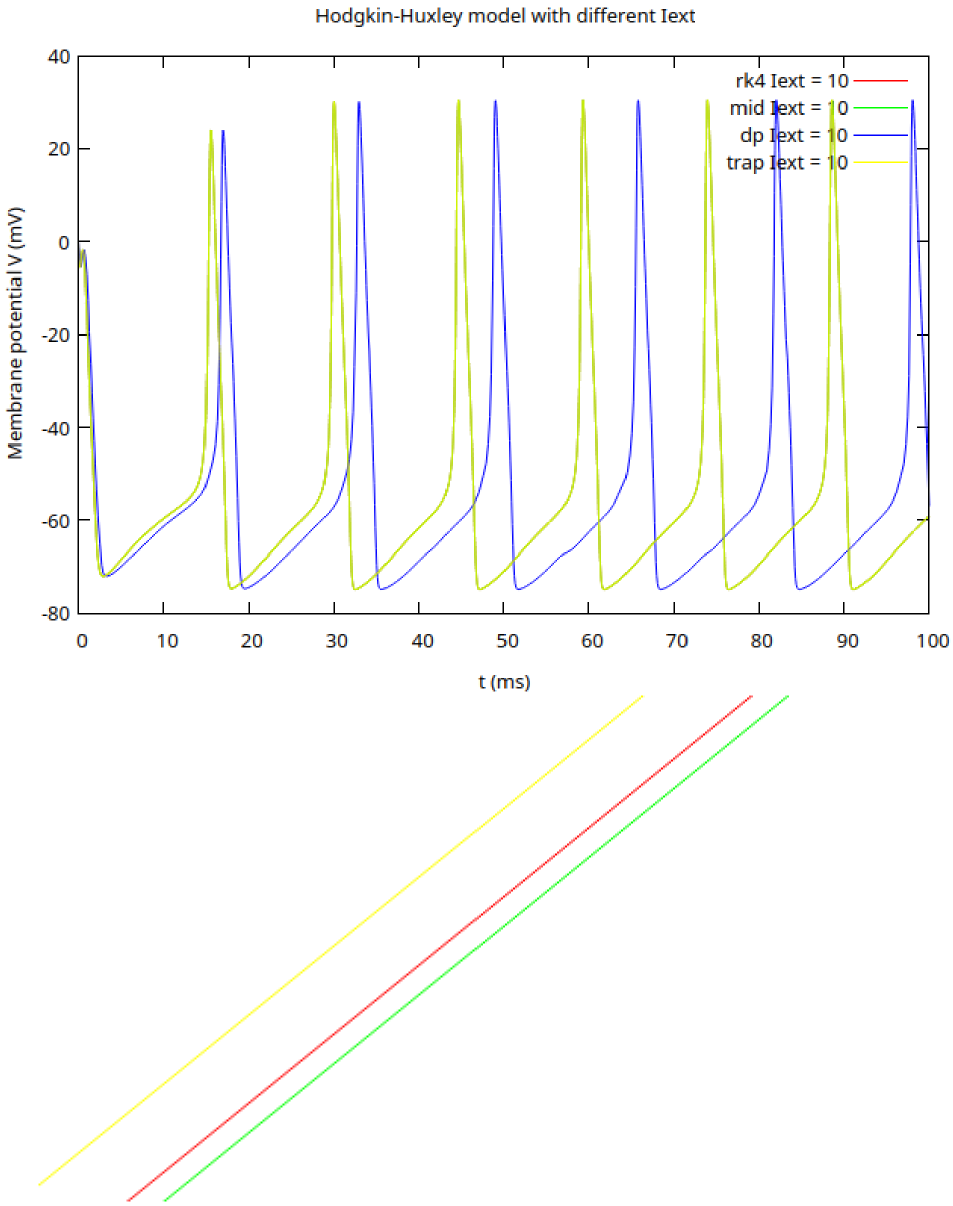
$$C_m\frac{dV_m}{dt}+I_{ion}=I_{ext}$$



Model(rk4) with another currents



Current-voltage picture



We can see that the Dormand–Prince method is the most useful. The Runge–Kutta 4, trapezoid, and midpoint methods have a local truncation error of a certain order**.**

RK4, trapezoid, and midpoint are fixed-step methods. They have finite local truncation errors ($O(h^5)$ for RK4, $O(h^3)$ for midpoint, $O(h^3)$ for trapezoid).